

Hamiltonian Generator Theoretical Background

0.1 Condensed Matter Physics

This category comprises canonical many-body quantum Hamiltonians fundamental to understanding emergent phenomena including quantum magnetism, phase transitions, and strongly correlated electron systems.

Heisenberg XYZ Model

The anisotropic Heisenberg model describes interacting spin systems with spatially dependent exchange coupling, providing a paradigmatic framework for quantum magnetism. The Hamiltonian for a one-dimensional chain with open boundary conditions is:

$$\hat{H}_{\text{XYZ}} = \sum_{i=0}^{n-2} (J_x \hat{\sigma}_i^x \hat{\sigma}_{i+1}^x + J_y \hat{\sigma}_i^y \hat{\sigma}_{i+1}^y + J_z \hat{\sigma}_i^z \hat{\sigma}_{i+1}^z) \quad (1)$$

where $\hat{\sigma}_i^\alpha$ ($\alpha \in \{x, y, z\}$) denote Pauli operators acting on qubit i , and n represents the chain length. The real-valued coupling constants $(J_x, J_y, J_z) \in \mathbb{R}^3$ determine the interaction anisotropy and thus the magnetic ordering.

This model exhibits rich phase behavior: when $J_x = J_y = J_z$, it reduces to the isotropic Heisenberg model; for $J_z \neq J_x = J_y$, it becomes the XXZ model showing a quantum phase transition; the fully anisotropic case with distinct couplings represents the most general spin nearest-neighbor interaction.

We generate unique Hamiltonians by uniformly sampling coupling constant tuples from the discrete grid $J_x, J_y, J_z \in [-3, 3]$ with spacing $\Delta J = 0.1$, ensuring comprehensive coverage of the parameter space including ferromagnetic ($J_k > 0$), antiferromagnetic ($J_k < 0$), and mixed interaction regimes.

Two-Dimensional Transverse-Field Ising Model

The transverse-field Ising model on a square lattice represents a paradigmatic system for studying quantum phase transitions driven by the competition between quantum fluctuations and classical ordering. The Hamiltonian is:

$$\hat{H}_{\text{TFIM}} = -J \sum_{\langle i,j \rangle} \hat{\sigma}_i^z \hat{\sigma}_j^z - h \sum_{k=0}^{n-1} \hat{\sigma}_k^x \quad (2)$$

where $\langle i, j \rangle$ denotes summation over nearest-neighbor pairs on a square lattice with open boundary conditions, $J \in \mathbb{R}^+$ is the Ising coupling strength, and $h \in \mathbb{R}^+$ represents the transverse field magnitude.

For n qubits arranged on a $\sqrt{n} \times \sqrt{n}$ square lattice (requiring n to be a perfect square), we construct the nearest-neighbor connectivity graph and generate Hamiltonians by sampling (J, h) pairs from a uniform grid where $J, h \in [0, 5]$ with spacing $\Delta = 0.1$.

Fermi-Hubbard Model

The one-dimensional Fermi-Hubbard model constitutes a fundamental framework for understanding strongly correlated fermionic systems, capturing the essential physics of competing kinetic and interaction energies. In second quantization, the Hamiltonian is:

$$\hat{H}_{\text{FH}} = -t \sum_{\langle i, j \rangle, \sigma} \left(\hat{c}_{i, \sigma}^\dagger \hat{c}_{j, \sigma} + \hat{c}_{j, \sigma}^\dagger \hat{c}_{i, \sigma} \right) + U \sum_{i=0}^{L-1} \hat{n}_{i, \uparrow} \hat{n}_{i, \downarrow} \quad (3)$$

where $\hat{c}_{i, \sigma}^\dagger$ ($\hat{c}_{i, \sigma}$) are fermionic creation (annihilation) operators for an electron with spin $\sigma \in \{\uparrow, \downarrow\}$ at site i , $\hat{n}_{i, \sigma} = \hat{c}_{i, \sigma}^\dagger \hat{c}_{i, \sigma}$ is the number operator, $t \in \mathbb{R}^+$ represents the hopping amplitude, $U \in \mathbb{R}^+$ is the on-site Coulomb repulsion strength, and L denotes the number of lattice sites.

The competition between kinetic energy (favoring delocalization) and Coulomb repulsion (favoring localization) gives rise to metal-insulator transitions and antiferromagnetic correlations.

Since quantum computers operate on qubits rather than fermionic modes, we employ the Jordan-Wigner transformation to map fermionic operators to Pauli operators:

$$\hat{c}_j^\dagger = \left(\prod_{k < j} \hat{\sigma}_k^z \right) \frac{\hat{\sigma}_j^x - i \hat{\sigma}_j^y}{2}, \quad \hat{c}_j = \left(\prod_{k < j} \hat{\sigma}_k^z \right) \frac{\hat{\sigma}_j^x + i \hat{\sigma}_j^y}{2} \quad (4)$$

This transformation preserves fermionic anticommutation relations while representing the system on $2L$ qubits (two per site for spin-up and spin-down orbitals). We generate instances by sampling (t, U) from a uniform grid where $t, U \in [0, 5]$ with spacing $\Delta = 0.1$.

0.2 Quantum Chemistry

Molecular electronic structure problems represent a primary application domain for variational quantum algorithms, with the objective of computing ground-state energies and properties of molecular systems. The electronic Hamiltonian in second quantization, employing the Born-Oppenheimer approximation, is:

$$\hat{H}_{\text{elec}} = \sum_{p,q} h_{pq} \hat{c}_p^\dagger \hat{c}_q + \frac{1}{2} \sum_{p,q,r,s} h_{pqrs} \hat{c}_p^\dagger \hat{c}_q^\dagger \hat{c}_r \hat{c}_s \quad (5)$$

where h_{pq} and h_{pqrs} are one- and two-electron integrals computed from molecular orbital basis functions.

We generate molecular Hamiltonians for the H_2 molecule using the minimal STO-3G basis set, which yields 4 spin-orbitals after Jordan-Wigner transformation. The internuclear distance R is varied from 0.5 Å to 4.97 Å in increments of 0.03 Å, producing unique molecular configurations spanning the dissociation curve from strongly bound to dissociated regimes.

Other larger molecules (LiH, BeH_2) have been implemented and tested on the provided code but not used on the final dataset.

0.3 Combinatorial Optimization

Combinatorial optimization problems on discrete solution spaces can be encoded as Ising Hamiltonians whose ground states correspond to optimal solutions. These problems are natural targets for quantum approximate optimization algorithms (QAOA) and serve as benchmarks for near-term quantum devices. The full derivations of the Max-Cut and Portfolio Hamiltonians are detailed in the Appendix 0.4.

Maximum Cut Problem

For an undirected weighted graph $G = (V, E, \mathbf{w})$ with vertex set V , edge set E , and edge weights $w_{ij} \geq 0$, the Maximum Cut problem seeks a bipartition $S \cup \bar{S} = V$ maximizing the total weight of edges crossing the cut. This NP-hard problem maps to a qubit Hamiltonian:

$$\hat{H}_{\text{MaxCut}} = \frac{1}{2} \sum_{(i,j) \in E} w_{ij} (\mathbb{I} - \hat{\sigma}_i^z \hat{\sigma}_j^z) \quad (6)$$

The ground state encodes the optimal partition: vertices i and j belong to opposite sets if $\langle \hat{\sigma}_i^z \hat{\sigma}_j^z \rangle < 0$ in the ground state. The constant term ensures positive eigenvalues.

For n -qubit problems, we generate random weighted graphs by: (i) sampling a random number of edges $m \in [n-1, n(n-1)/2]$ to ensure connectivity, (ii) selecting m unique vertex pairs uniformly from all $\binom{n}{2}$ possibilities, (iii) assigning edge weights $w_{ij} \sim \mathcal{U}(0.1, 5.0)$. This procedure is repeated to generate diverse graph topologies and weight distributions.

Portfolio Optimization

Financial portfolio optimization balances expected returns against risk (variance) when selecting assets. For n assets with expected returns $\mathbf{r} = (r_1, \dots, r_n)$:

$$\hat{H}_{\text{Portfolio}} = - \sum_{i=0}^{n-1} r_i \hat{\sigma}_i^z + \lambda \sum_{i,j=0}^{n-1} \Sigma_{ij} \hat{\sigma}_i^z \hat{\sigma}_j^z \quad (7)$$

Where $z_i \in \{-1, +1\}$ indicates asset selection and $\lambda > 0$ is the risk aversion parameter.

We generate random problem instances by sampling the expected returns $r_i \sim \mathcal{U}(-0.1, 0.2)$ representing realistic return distributions and keeping the risk penalty fixed at $\lambda = 0.5$.

0.4 Random Matrix Theory

To mitigate potential biases toward physically structured Hamiltonians and ensure model robustness across the full operator space, we include Hamiltonians sampled from random matrix ensembles. These provide unstructured test cases without inherent symmetries or locality constraints.

Gaussian Orthogonal Ensemble (GOE)

The GOE comprises real symmetric matrices with entries drawn independently from Gaussian distributions, modeling time-reversal invariant quantum systems with no additional symmetries. For n qubits, the operator space dimension is 4^n , spanned by Pauli strings $\{\mathbb{I}, \hat{\sigma}^x, \hat{\sigma}^y, \hat{\sigma}^z\}^{\otimes n}$.

To construct Hermitian Hamiltonians from this ensemble while respecting reality constraints, we sample coefficients only for Pauli strings containing

an even number of $\hat{\sigma}^y$ operators (ensuring real matrix elements):

$$\hat{H}_{\text{GOE}} = \sum_{\substack{\mathbf{P} \in \mathcal{P}_n \\ |\{j: P_j=Y\}| \text{ even}}} c_{\mathbf{P}} \hat{\mathbf{P}}, \quad c_{\mathbf{P}} \sim \mathcal{N}(0, \sigma^2) \quad (8)$$

where $\mathcal{P}_n$ denotes the set of all n -qubit Pauli strings and σ controls the energy scale.

Sparse Random Hamiltonians

Physical Hamiltonians typically exhibit locality—interactions involve only small subsets of qubits (k -local terms). We generate sparse Hamiltonians by:

$$\hat{H}_{\text{sparse}} = \sum_{\mathbf{P} \in \mathcal{S}} c_{\mathbf{P}} \hat{\mathbf{P}} \quad (9)$$

where $\mathcal{S} \subset \mathcal{P}_n$ is a randomly selected subset of Pauli strings with at most k non-identity operators, and $|\mathcal{S}|$ is determined by density parameter $\rho \in (0, 1]$.

We use $k = 2$ (two-local terms), $\rho = 0.1$ (10% of possible terms), and $c_{\mathbf{P}} \sim \mathcal{N}(0, 1)$, yielding Hamiltonians structurally similar to quantum chemistry and condensed matter problems but with random interaction strengths.

Power-Law Hamiltonians

Long-range interacting systems, including trapped ions and Rydberg atoms, exhibit coupling strengths decaying as power laws of spatial separation. We model this with all-to-all two-qubit interactions:

$$\hat{H}_{\text{PL}} = \sum_{0 \leq i < j < n} \frac{c_{ij}}{|i - j|^\alpha} \hat{\sigma}_i^{P_i} \hat{\sigma}_j^{P_j} \quad (10)$$

where $P_i, P_j \in \{x, y, z\}$ are randomly selected Pauli types, $c_{ij} \sim \mathcal{N}(0, 1)$ are random coupling constants, and $\alpha > 0$ controls interaction range ($\alpha = 2$ for dipole-dipole interactions, $\alpha = 3$ for van der Waals).

This multi-domain approach ensures the training dataset exhibits robust statistical properties essential for generalization. It guarantees structural diversity, encompassing Hamiltonians that span the range from local (nearest-neighbor) to non-local (all-to-all) interaction topologies. Furthermore, the

dataset features wide symmetry variation, from highly symmetric to completely asymmetric (e.g., random GOE) systems. By covering a broad spectrum of spectral characteristics, including different gap structures and entanglement properties we ensure the model could gains proficiency across major application domains for near-term quantum algorithms.

Appendix

1 Hamiltonian Encoding Of Combinatorial Optimization Problems

This appendix provides detailed derivations for encoding classical combinatorial optimization problems as quantum Hamiltonians, bridging the gap between discrete optimization and quantum computing formulations.

Binary-to-Spin Variable Transformation

To transform binary variables $x_i \in \{0, 1\}$ to spin variables $z_i \in \{-1, +1\}$:

$$x_i = \frac{1 - z_i}{2} \quad (11)$$

This transformation maps:

$$x_i = 0 \Rightarrow z_i = +1 \quad (\text{spin up}, |\uparrow\rangle = |0\rangle) \quad (12)$$

$$x_i = 1 \Rightarrow z_i = -1 \quad (\text{spin down}, |\downarrow\rangle = |1\rangle) \quad (13)$$

Spin-to-Pauli Operator Mapping

Quantum mechanically, spin variables map to Pauli-Z operators:

$$z_i \rightarrow \hat{\sigma}_i^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}_i \quad (14)$$

With computational basis eigenvalues:

$$\hat{\sigma}_i^z |0\rangle_i = +1 |0\rangle_i \quad (z_i = +1) \quad (15)$$

$$\hat{\sigma}_i^z |1\rangle_i = -1 |1\rangle_i \quad (z_i = -1) \quad (16)$$

Thus we can transform a classical objective function into a quantum Hamiltonian operator acting on the computational basis.

E.1.1 Maximum Cut Problem

Problem Statement

Given an undirected weighted graph $G = (V, E, \mathbf{w})$ where:

- $V = \{0, 1, \dots, n-1\}$ is the vertex set (mapped to qubits)
- $E \subseteq V \times V$ is the edge set
- $w_{ij} > 0$ is the weight of edge $(i, j) \in E$

The goal is to partition V into two disjoint sets S and $\bar{S} = V \setminus S$ such that the total weight of edges crossing the cut is maximized:

$$\text{MaxCut}(G) = \max_{S \subseteq V} \sum_{\substack{(i,j) \in E \\ i \in S, j \in \bar{S}}} w_{ij} \quad (17)$$

Binary Variable Encoding

Encode the partition using binary indicator variables:

$$x_i = \begin{cases} 1 & \text{if vertex } i \in S \\ 0 & \text{if vertex } i \in \bar{S} \end{cases} \quad (18)$$

An edge (i, j) crosses the cut if and only if $x_i \neq x_j$, which occurs when:

$$x_i(1 - x_j) + (1 - x_i)x_j = x_i + x_j - 2x_ix_j = 1 \quad (19)$$

Otherwise the equation is equal to 0.

Therefore, the objective function can be written as:

$$f(\mathbf{x}) = \sum_{(i,j) \in E} w_{ij}(x_i + x_j - 2x_ix_j) \quad (20)$$

Sum w_{ij} only if the nodes belong to different subsets.

Spin Variable Transformation

Applying the transformation $x_i = (1 - z_i)/2$:

$$x_i + x_j - 2x_i x_j = \frac{1 - z_i}{2} + \frac{1 - z_j}{2} - 2 \cdot \frac{(1 - z_i)(1 - z_j)}{4} \quad (21)$$

$$= \frac{2 - z_i - z_j}{2} - \frac{(1 - z_i)(1 - z_j)}{2} \quad (22)$$

$$= \frac{2 - z_i - z_j - 1 + z_i + z_j - z_i z_j}{2} \quad (23)$$

$$= \frac{1 - z_i z_j}{2} \quad (24)$$

Thus the MaxCut objective in spin variables is:

$$f(\mathbf{z}) = \sum_{(i,j) \in E} w_{ij} \frac{1 - z_i z_j}{2} = \frac{1}{2} \sum_{(i,j) \in E} w_{ij} (1 - z_i z_j) \quad (25)$$

Quantum Hamiltonian

Since we seek to *maximize* $f(\mathbf{z})$, the corresponding minimization Hamiltonian is:

$$\hat{H}_{\text{MaxCut}} = -f(\mathbf{z}) = -\frac{1}{2} \sum_{(i,j) \in E} w_{ij} (1 - z_i z_j) \quad (26)$$

Replacing spin variables with Pauli operators $z_i \rightarrow \hat{\sigma}_i^z$:

$$\boxed{\hat{H}_{\text{MaxCut}} = -\frac{1}{2} \sum_{(i,j) \in E} w_{ij} (\mathbb{I} - \hat{\sigma}_i^z \hat{\sigma}_j^z)} \quad (27)$$

Alternatively, absorbing the negative sign and constant:

$$\hat{H}_{\text{MaxCut}} = \frac{1}{2} \sum_{(i,j) \in E} w_{ij} (\mathbb{I} - \hat{\sigma}_i^z \hat{\sigma}_j^z) \quad (28)$$

where the sign convention depends on whether the quantum algorithm minimizes or maximizes the expectation value.

E.1.2 Portfolio Optimization

Problem Statement

Consider n financial assets with:

- Expected returns: $\mathbf{r} = (r_1, \dots, r_n) \in \mathbb{R}^n$
- Covariance matrix: $\Sigma \in \mathbb{R}^{n \times n}$ (symmetric, positive semi-definite)
- Risk aversion parameter: $\lambda > 0$

The Markowitz mean-variance portfolio optimization problem is:

$$\max_{\mathbf{x} \in \{0,1\}^n} \left[\sum_{i=0}^{n-1} r_i x_i - \lambda \sum_{i,j=0}^{n-1} \Sigma_{ij} x_i x_j \right] \quad (29)$$

where $x_i = 1$ indicates asset i is included in the portfolio.

Spin Variable Transformation

Substituting $x_i = (1 - z_i)/2$:

Linear term (expected return):

$$\sum_i r_i x_i = \sum_i r_i \frac{1 - z_i}{2} = \frac{1}{2} \sum_i r_i (1 - z_i) \quad (30)$$

Quadratic term (risk):

$$\sum_{i,j} \Sigma_{ij} x_i x_j = \sum_{i,j} \Sigma_{ij} \frac{(1 - z_i)(1 - z_j)}{4} \quad (31)$$

$$= \frac{1}{4} \sum_{i,j} \Sigma_{ij} (1 - z_i - z_j + z_i z_j) \quad (32)$$

Combined Objective

The full objective becomes:

$$f(\mathbf{z}) = \frac{1}{2} \sum_i r_i (1 - z_i) - \frac{\lambda}{4} \sum_{i,j} \Sigma_{ij} (1 - z_i - z_j + z_i z_j) \quad (33)$$

$$= \underbrace{\frac{1}{2} \sum_i r_i - \frac{\lambda}{4} \sum_{i,j} \Sigma_{ij}}_{\text{constant}} - \frac{1}{2} \sum_i r_i z_i + \frac{\lambda}{4} \sum_{i,j} \Sigma_{ij} (z_i + z_j - z_i z_j) \quad (34)$$

Dropping constants (they don't affect optimization):

$$f(\mathbf{z}) \propto -\frac{1}{2} \sum_i r_i z_i + \frac{\lambda}{4} \sum_{i,j} \Sigma_{ij} (z_i + z_j - z_i z_j) \quad (35)$$

Simplification

The linear terms in the double sum:

$$\sum_{i,j} \Sigma_{ij} z_i = \sum_i z_i \sum_j \Sigma_{ij} \quad (36)$$

For simplicity, assuming we can absorb these into the single-qubit terms, the final Hamiltonian (converting maximization to minimization) is:

$$\hat{H}_{\text{Portfolio}} = - \sum_{i=0}^{n-1} r_i \hat{\sigma}_i^z + \lambda \sum_{i,j=0}^{n-1} \Sigma_{ij} \hat{\sigma}_i^z \hat{\sigma}_j^z \quad (37)$$